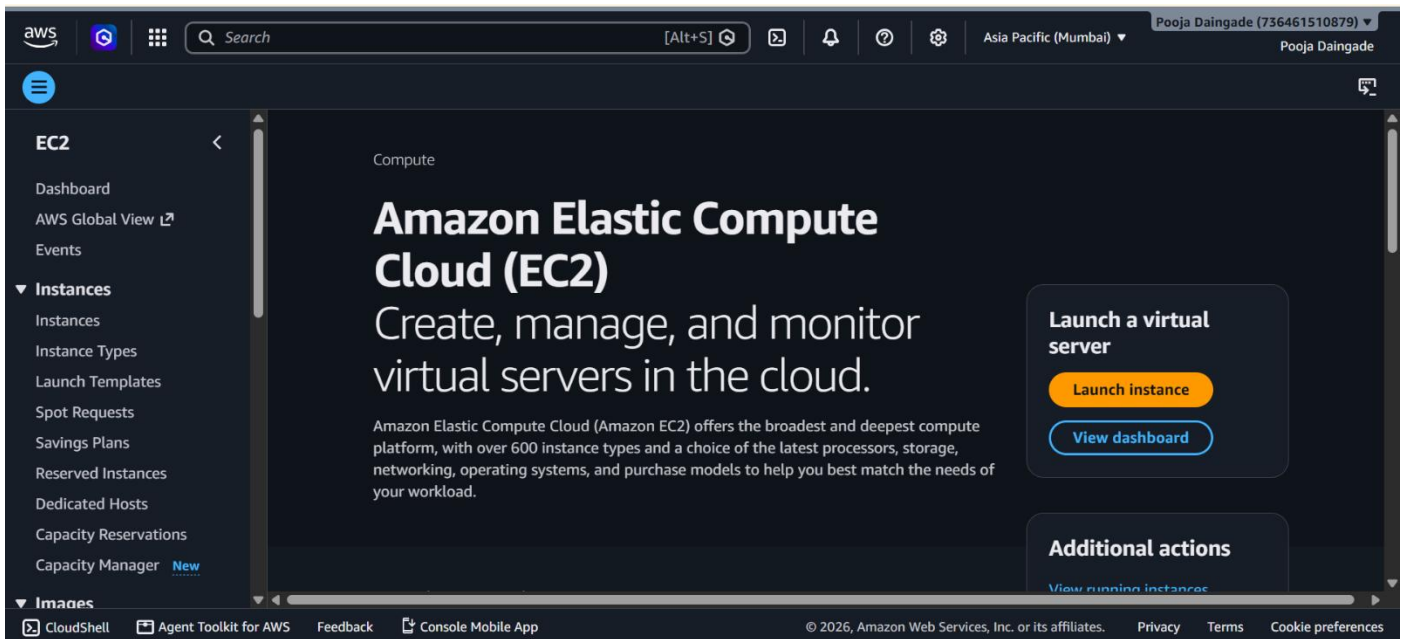


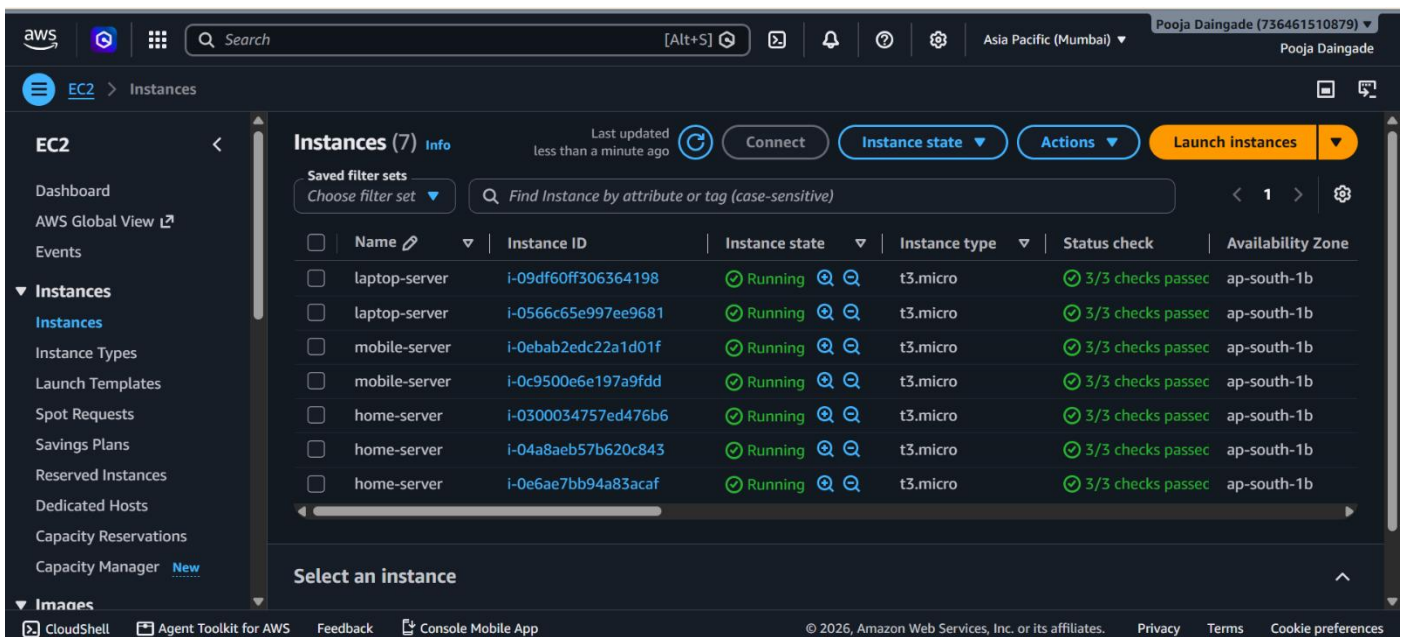
AWS Hybrid Load Balancing using Application Load Balancer (ALB) and Network Load Balancer (NLB)

By: Pooja Daingade

1. Sign in to the AWS Management Console with credentials and open the EC2 service home page.



2. Launch a total of 7 Amazon EC2 instances, including:
 - 3→Home servers
 - 2→Mobile servers
 - 2→Laptop servers



3. Create Target Groups (for ALB): Home Target Group Mobile Target Group Laptop Target Group

Select Target Type as **Instances**.

Step 1
Create target group
Step 2 - recommended
Register targets
Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation."

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: **ALB** **NLB** **GWLB**

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: **ALB** **NLB** **GWLB**

☐ **Lambda function**
Supports load balancing to a single

☐ **Application Load Balancer**
Allows use of static IP addresses and

4. Enter the target group name as **home-tg**, choose **HTTP** as the protocol with **Port 80**, keep **IPv4** as the IP address type, and proceed with the default settings to create the target group.

For the other target groups, just change the name:

Mobile Target Group: Enter the target group name as **mobile-tg**.

Laptop Target Group: Enter the target group name as **laptop-tg**.

Target group name
Name must be unique per Region per AWS account.

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 7/32

Protocol
Protocol for communication between the load balancer and targets.

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

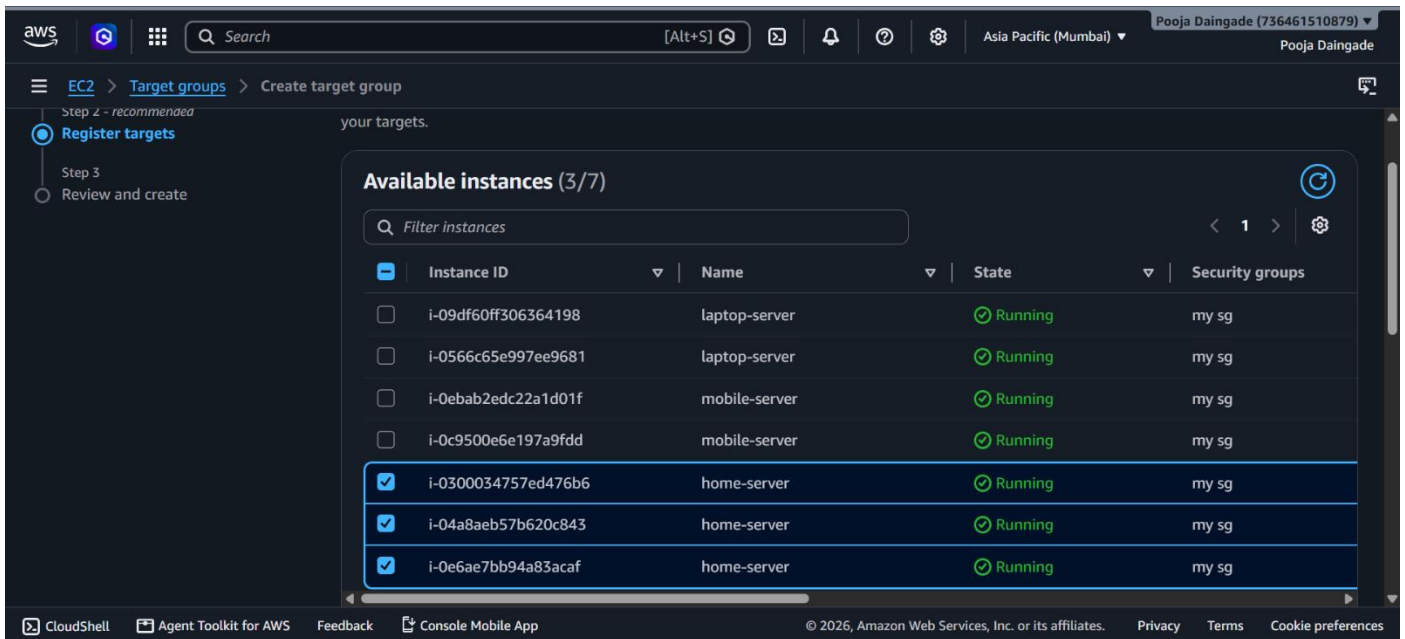
☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the

5. Select the 3→**Home Servers** and click **Include as pending below** to register them with the Home Target Group. Review the selected instances and click Create Target Group to complete the configuration.

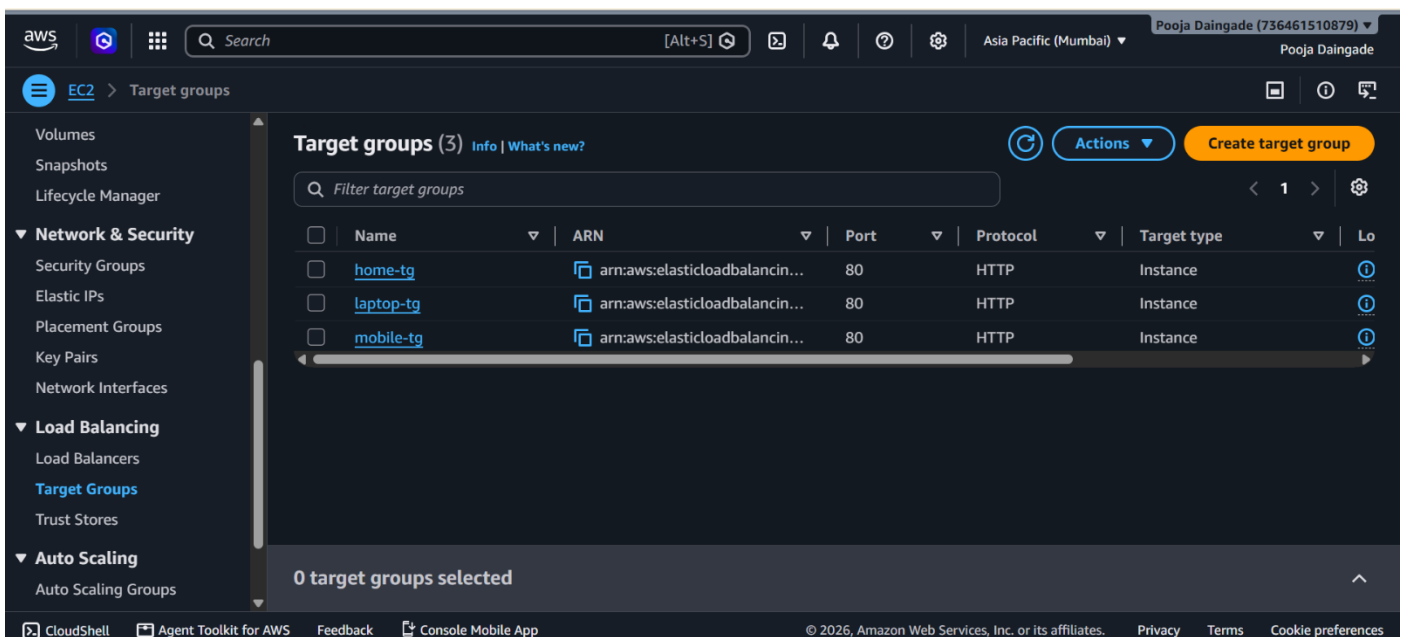
For the other target groups, simply change the server names:

Mobile Target Group: Select the 2→**Mobile Servers** and click **Include as pending below**.

Laptop Target Group: Select the 2→**Laptop Servers** and click **Include as pending below**.



6. Successfully created 3 Target Groups: home-tg
mobile-tg
laptop-tg.



- Click **Create Load Balancer** and select **Application Load Balancer**. Enter the load balancer name as **ALB**, choose **Internet-facing** as the scheme, select **IPv4** as the IP address type.

Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

► **How Application Load Balancers work**

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**
• Serves internet-facing traffic.

☐ **Internal**
• Serves internal traffic.

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- Scroll to the **Availability Zones and subnets** section, select the all **Availability Zones** for high availability.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **ap-south-1a (aps1-az1)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-089645b44f435f232
172.31.32.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ **ap-south-1b (aps1-az3)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-079558261318972e8
172.31.0.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ **ap-south-1c (aps1-az2)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

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9. Next, choose the **Security Group** to allow inbound and outbound traffic, configure the **HTTP listener on Port 80**.

The screenshot shows the 'Security groups' section of the 'Create Application Load Balancer' wizard. It includes a description of security groups and a dropdown menu to select up to 5 security groups. A selected group, 'my sg (sg-0319f7c4ca292adb2)', is shown with its inbound and outbound rule counts. Below this is the 'Listeners and routing' section, which shows a listener named 'HTTP:80' with Protocol set to 'HTTP' and Port set to '80'. A 'Remove' button is visible for the listener.

10. Configure the Default Action by selecting **Forward to target groups** and choosing **home-tg**. Keep the default settings and continue to create the Application Load Balancer.

The screenshot shows the 'Default action' section of the 'Create Application Load Balancer' wizard. It includes a description of the default action and three radio button options for the routing action: 'Forward to target groups' (selected), 'Redirect to URL', and 'Return fixed response'. Below the routing action is the 'Forward to target group' section, which includes a dropdown for the target group (set to 'home-tg'), a dropdown for the protocol (set to 'HTTP'), a weight input field (set to '1'), and a percent input field (set to '100%').

Verify that the **Application Load Balancer (ALB)** is created successfully and its status changes to **Active**.

The screenshot shows the 'Load balancers' page in the AWS Management Console. It includes a search bar, a table of load balancers, and a sidebar with navigation links. The table shows one load balancer, 'ALB', with a status of 'Active'. The sidebar includes links to 'Volumes', 'Snapshots', 'Lifecycle Manager', 'Network & Security', 'Load Balancing', and 'Auto Scaling'.

Name	State	Type	Scheme	IP address type	VPC ID
ALB	Active	application	Internet-facing	IPv4	vpc-083ff097

11. Configure Path-Based Routing: Add rule→Conditions→Path.

Path = /mobile/*

Conditions (1 value) [Info](#) [Rule limits](#)

Define 1-5 condition values. Additional conditions can't be added once the limit is reached.

▼ **Path (value) =** [Remove](#)

Match pattern type

☒ **Value matching**
Match with glob syntax, using `*` and `?` as wildcards.

☐ **Regex matching**
Match with regex syntax.

Path condition value
Case sensitive.

=

Valid characters are a-z, A-Z, 0-9 and [special characters](#). Path must be 1-128 characters.

[+ Add OR condition value](#)

[Add condition](#) ▼

12. Then choose **Forward to target group** and select **mobile-tg**. Next Set the rule priority to 1 for the /mobile/* path rule

Routing action

☒ **Forward to target groups** ☐ Redirect to URL ☐ Return fixed response

Forward to target group [Info](#)
Choose a target group and specify routing weight or [create target group](#).

Target group

Select a target group

home-tg	HTTP
laptop-tg	HTTP
mobile-tg	HTTP

You can add up to 4 more target groups.

Set rule priority [Info](#)

Each rule has a priority. The default rule is evaluated last. You can change the priority of a non-default rule at any time. You can't change the priority of the default rule.

► **Listener details: HTTP:80**

Listener rules (2) [Info](#) [Rule limits](#) [Reset priorities](#) [Add gap between priorities](#) ▼

Traffic received by the listener is routed according to the default action and any additional rules. Rules are evaluated in priority order from the lowest value to the highest value.

Priority	Name tag	Conditions (If)	Transforms
1	-	Path (value) = /mobile/*	-
Priority value must be 1-50,000.			
Last (default)	Default	If no other rule applies	-

13. Configure Path-Based Routing: Add rule→Conditions→Path.

Path = /laptop/*

Conditions (1 value) [Info](#) [Rule limits](#)

Define 1-5 condition values. Additional conditions can't be added once the limit is reached.

▼ **Path (value) =** [Remove](#)

Match pattern type

☒ **Value matching**
Match with glob syntax, using `*` and `?` as wildcards.

☐ **Regex matching**
Match with regex syntax.

Path condition value
Case sensitive.

=

Valid characters are a-z, A-Z, 0-9 and [special characters](#). Path must be 1-128 characters.

[+ Add OR condition value](#)

14. Then choose **Forward to target group** and select **laptop-tg**. Next Set the rule priority to 2 for the /laptop/* path rule

Actions [Info](#)

Requests matching all rule conditions route according to the rule actions.

Routing action

☒ **Forward to target groups** ☐ Redirect to URL ☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

Select a target group

home-tg	Target type: Instance, IPv4 Target stickiness: Off	HTTP
laptop-tg	Target type: Instance, IPv4 Target stickiness: Off	HTTP
mobile-tg	Target type: Instance, IPv4 Target stickiness: Off	HTTP

Set rule priority [Info](#)

Each rule has a priority. The default rule is evaluated last. You can change the priority of a non-default rule at any time. You can't change the priority of the default rule.

► **Listener details: HTTP:80**

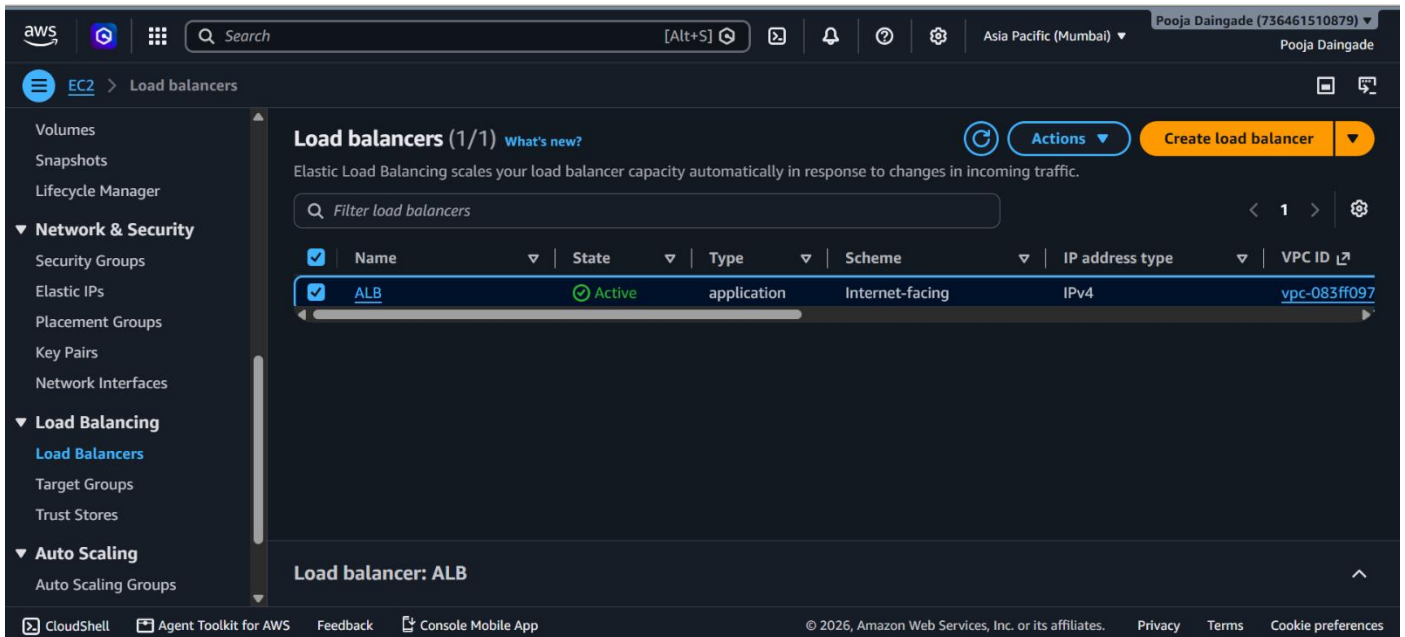
Listener rules (3) [Info](#) [Rule limits](#) [Reset priorities](#) [Add gap between priorities](#)

Traffic received by the listener is routed according to the default action and any additional rules. Rules are evaluated in priority order from the lowest value to the highest value.

Priority	Name tag	Conditions (If)	Transforms
1	-	Path (value) = /mobile/*	-
2	-	Path (value) = /laptop/*	-

Priority value must be 1-50,000.

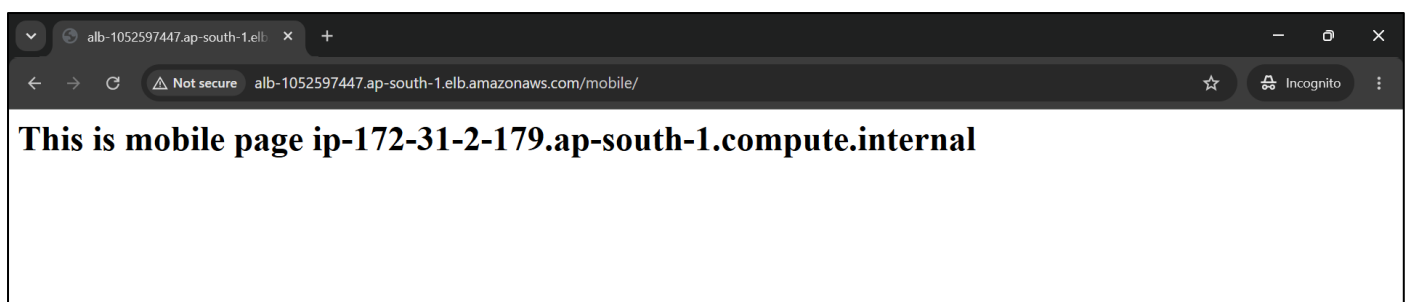
15. Successfully created the Application Load Balancer (ALB).



16. Test ALB: http://<ALB-DNS>/



http://<ALB-DNS>/mobile/



http://<ALB-DNS>/laptop/



17. Create Target Group for NLB.

IMP Select Target Type as → **Application Load Balancer**

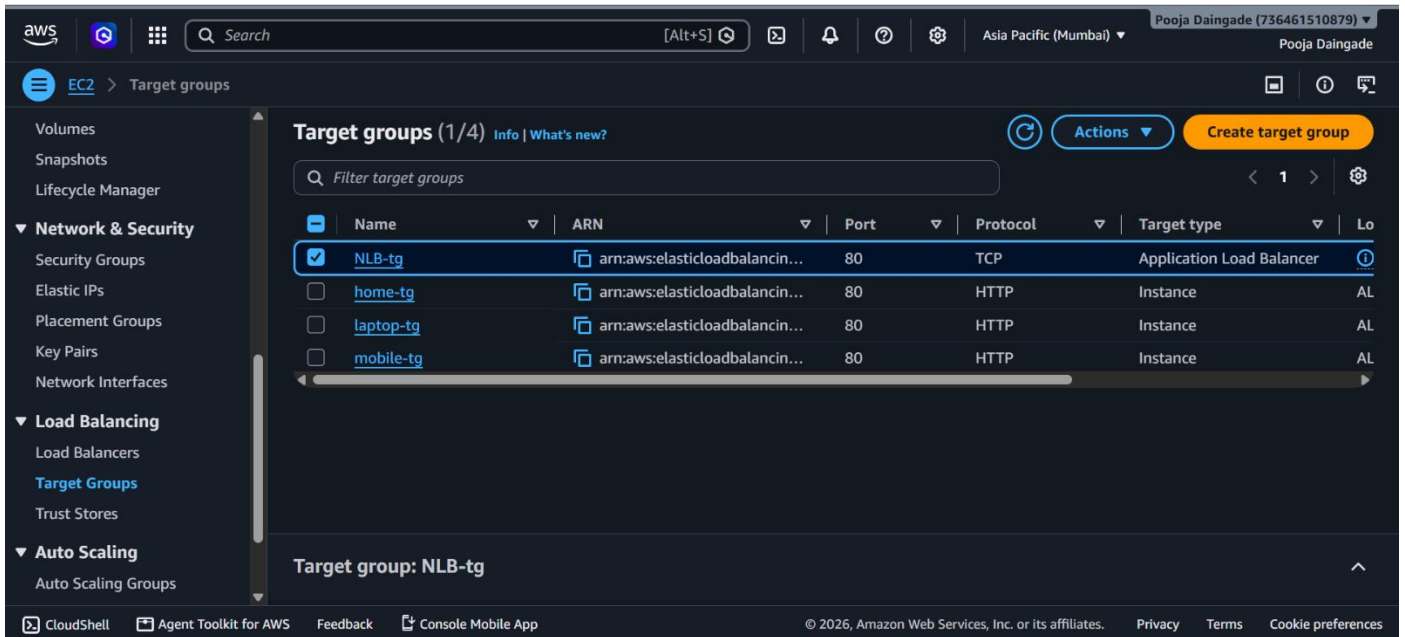
The screenshot shows the AWS Management Console interface for creating a target group. The breadcrumb navigation indicates the path: EC2 > Target groups > Create target group. The left sidebar shows the steps: Step 1: Create target group (selected), Step 2 - recommended: Register targets, and Step 3: Review and create. The main content area is titled 'Create target group' and includes a description: 'A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.' Below this is the 'Settings - immutable' section, which states: 'Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.' The 'Target type' section asks to 'Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.' There are four options: 'Instances' (suitable for ALB, NLB, GWLB), 'IP addresses' (suitable for ALB, NLB, GWLB), 'Lambda function' (suitable for ALB), and 'Application Load Balancer' (selected, suitable for ALB, NLB, GWLB). The 'Application Load Balancer' option is highlighted with a blue border and a blue radio button.

18. Enter the target group name as **NLB-tg.**, choose **TCP** as the protocol with **Port 80**, Select **ALB** which we created earlier.

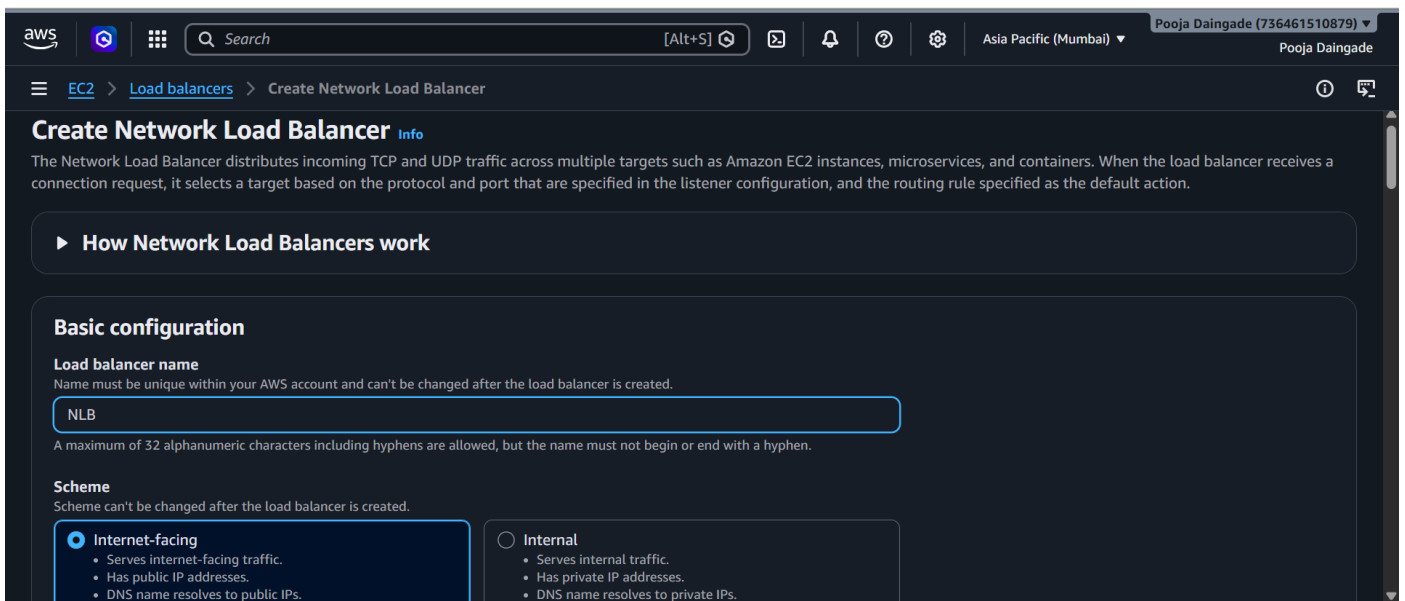
The screenshot shows the 'Create target group' page with the following fields filled out: 'Target group name' is 'NLB-tg', 'Protocol' is 'TCP', 'Port' is '80', and 'VPC' is 'vpc-083ff097c6113708f'. The 'VPC' field also shows the CIDR block '172.31.0.0/16'. There is a 'Create VPC' link next to the VPC field. The page also includes a 'Feedback' link and a 'Console Mobile App' link.

The screenshot shows the 'Register targets - recommended' page. The breadcrumb navigation indicates the path: EC2 > Target groups > Create target group. The left sidebar shows the steps: Step 1: Create target group, Step 2 - recommended: Register targets (selected), and Step 3: Review and create. The main content area is titled 'Register targets - recommended' and includes a description: 'This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.' Below this is the 'Register Application Load Balancer' section, which states: 'You can specify a single Application Load Balancer as the target.' There are two options: 'Register now' (selected) and 'Register later'. The 'Port' section asks: 'The Application Load Balancer target will receive traffic on this port and must have a listener on the same port.' There are two options: 'Use the target group port 80' (selected) and 'Specify an override port'. The 'Application Load Balancer' section asks: 'Choose an Application Load Balancer from the list, or create a new one and refresh the list to select it.' The 'ALB' option is selected in the dropdown menu.

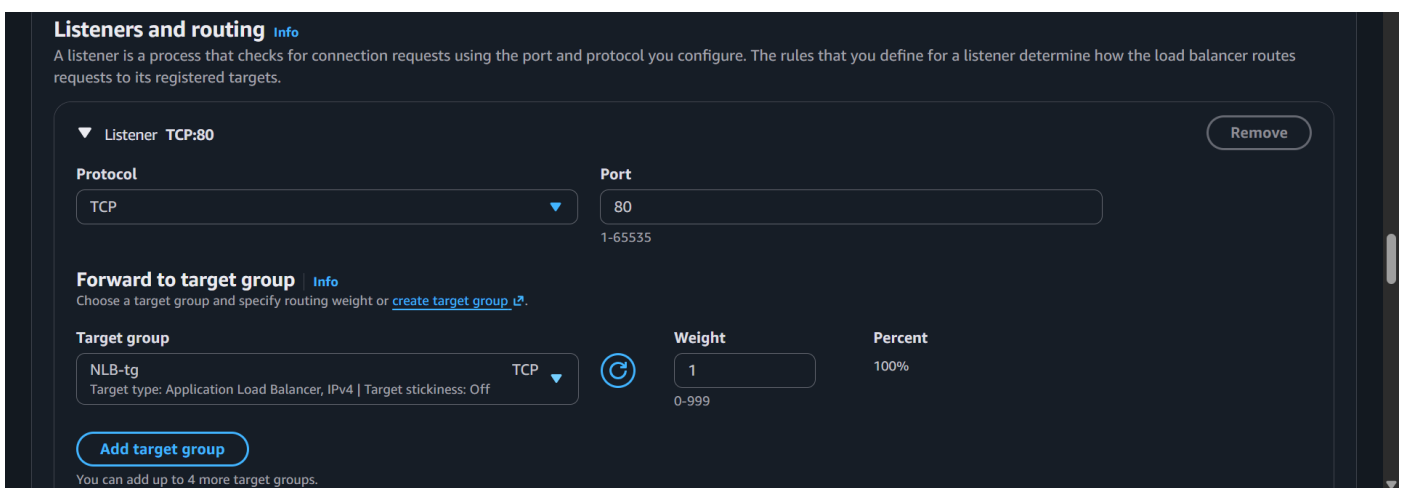
19.Target Group for NLB is created.



20.Create Network Load Balancer (NLB). Enter the load balancer name as **NLB**. Choose **Internet-facing** as the scheme.



Forward to target group- select-NLB-tg



21. Successfully created the Network Load Balancer (NLB).

The screenshot shows the AWS Management Console interface. The left sidebar contains navigation links for Volumes, Snapshots, Lifecycle Manager, Network & Security, Load Balancing, and Auto Scaling. The main content area displays the 'Load balancers' page with a table listing two load balancers: 'NLB' (Network Load Balancer) and 'ALB' (Application Load Balancer). The 'NLB' is selected, and its details are shown at the bottom of the console.

Name	State	Type	Scheme	IP address type	VPC ID
NLB	Active	network	Internet-facing	IPv4	vpc-083ff097
ALB	Active	application	Internet-facing	IPv4	vpc-083ff097

Load balancer: NLB

22. Copy the DNS of NLB and hit it on chrome.

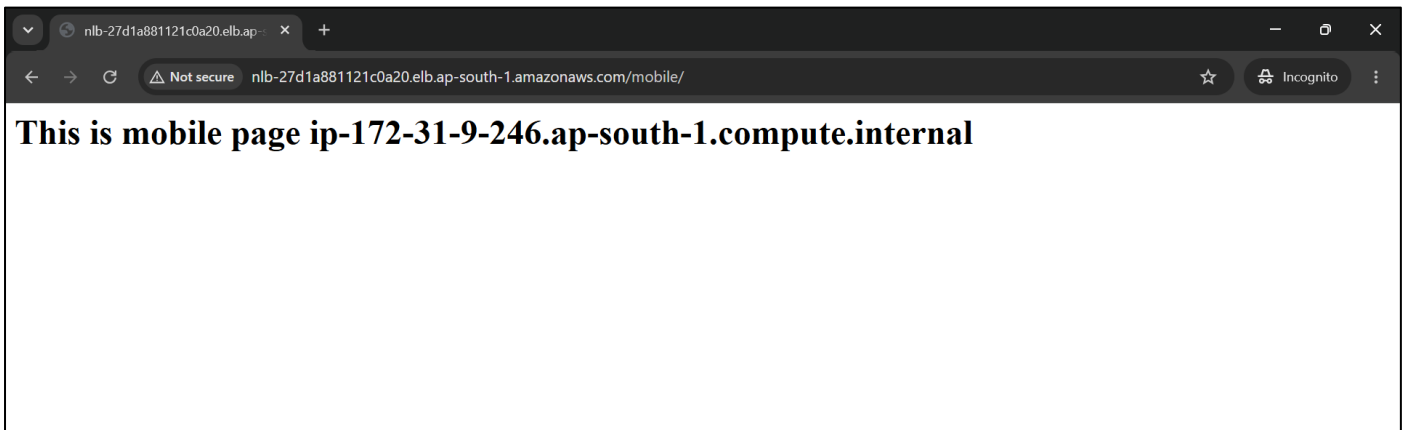
The screenshot shows the 'Details' page for the Network Load Balancer (NLB). The page displays various details including the load balancer type, status, scheme, VPC, availability zones, date created, load balancer ARN, and DNS name.

Details			
Load balancer type Network	Status Active	VPC vpc-083ff097c6113708f	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZVDDRQ8TROA	Availability Zones subnet-089645b44f435f232 ap-south-1a (aps1-az1) subnet-079558261318972e8 ap-south-1b (aps1-az3) subnet-09e859df578f607e6 ap-south-1c (aps1-az2)	Date created July 20, 2026, 17:25 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:ap-south-1:736461510879:loadbalancer/net/NLB/27d1a881121c0a20		DNS name Info NLB-27d1a881121c0a20.elb.ap-south-1.amazonaws.com (A Record)	

Home page



Mobile page



Laptop page

